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Energy-efficient task scheduling and physiological assessment in disaster management using UAV-assisted networks

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ABSTRACT

Internet of Things (IoT) and unmanned aerial vehicles (UAVs) together can significantly enhance the performance of disaster management systems. UAVs can collect massive heterogeneous data from disaster-affected areas using fifth-generation (5G)/beyond 5G networks and this data can be analyzed to get the information required by the first responders such as marking the boundary of the affected area, identifying the infrastructure damaged and the roads blocked, and the health situation of people living in that area. This paper presents an overview of different platforms (UAVs-based, IoT-based, and IoT, coupled with UAVs) for disaster management. We propose an energy-efficient task scheduling scheme for data collection by UAVs from the ground IoT network. The focus is to optimize the path taken by the UAVs to minimize energy consumption. We also analyze the vital signs data collected by UAVs for people in disaster-affected areas and apply the decision tree classification algorithm to determine their health risk status. The risk status will enable the first responders to decide the areas which need the most immediate help Simulation results compare the effectiveness of our proposed scheduling scheme with the traditional approach used for data collection. Also, we present the results of our predicted risk status compared with the risk status calculated through the National Early Warning Score 2 (NEWS2) method.

1. Introduction

Unmanned Aerial Vehicles (UAVs) have countless applications in disaster management, agriculture, surveillance, and numerous public services due to their mobility, flexibility, and ability to fly without a pilot [1]. Such applications place high requirements on communication in terms of reliability, energy consumption, throughput, and latency. Fifth-generation (5G) and beyond 5G networks have the potential to meet these requirements [2]. Internet of Things (IoT) platform for disaster management using UAVs can save precious human lives by aiding the emergency rescue team in the disaster area. UAVs can be launched strategically and automatically during search and rescue (SAR) operations [3]. Wireless networks comprising of UAVs can offer limited connectivity to disaster-struck regions where landline infrastructure has been destroyed. Focusing on the various aspects of wireless connectivity in UAVs, authors in [4] presented UAV based communication as a key enabler for 5G in public safety networks. UAVs can provide communication support for the ground IoT network to perform safety tasks in SAR operations. Further, UAVs can be equipped with remotely controllable IoT sensors and actuators. We can also analyze the large amount of data collected from ground IoT network and UAVs, and deduce vital information in disaster scenarios. For example, after the earthquake in Nepal in 2015, UAVs were used for aerial mapping of the affected areas, which provided a good understanding of the needs on the ground [5].

The deployment of the sensor network using UAVs for disaster management is presented in [6]. In addition, UAVs are used for capturing and evaluating the terrain of the disaster-affected area. The recent development of UAVs-assisted disaster management systems is presented in [3]. Authors presented architectures for different types of disasters based on the interaction between ground IoT network and UAVs. Further, three disaster stages (i.e., pre-disaster preparedness, disaster assessment, and disaster response and recovery) are considered for deployment scenarios for IoT network and UAVs. However, for efficient IoT platform using UAVs, both centralized and distributed platforms should be aware of their heterogeneous contextual information, e.g., their flying path, power status, equipment specifications, etc. [7]. Developing an IoT platform for disaster management using UAVs has

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Table 1
NEWS scoring system from Royal College of Physicians [10].

Score	HR	S_pO_2	NBP	awRR	T
3	≤40	≤91	≤90	≤8	≤35.0
2		92–93	91–100		
1	41–50	94–95	101–110	9–11	35.1–36.0
0	51–90	≥96	111–219	12–20	36.1–38.0
1	91–110				38.1–39.0
2	111–130			21–24	≥39.1
3	≥130		≥131	≥ 25	

several real challenges, including self-organized maintenance, communication between UAVs and with ground IoT network, designing an architecture to connect heterogeneous devices, and energy efficiency in operating the UAVs along with the connected devices [3].

The data collected by UAVs can be analyzed to classify clusters according to the number of people at risk. To find the risk factor of the cluster, we need to find the individual risk status of people in the disaster situation. The risk status can be predicted based on vital signs (physiological parameters) of people in the disaster-affected area, which will be an indication of events such as death, cardiac arrest, and intensive care unit (ICU) admission [8]. These vital signs can include heart rate (HR), blood oxygen saturation (S_pO_2), systolic blood pressure (NBP), respiratory rate (awRR), and temperature. A simple additive approach is among the early efforts to find the overall risk status of the person based on vital signs [9]. However, this approach has some limitations mentioned in [9]. This risk scoring system can be defined according to the National Early Warning Score (NEWS) [10] developed in 2012 (updated to version 2 in 2017) by the Royal College of Physicians (RCP), which is shown in Table 1. Then the score is classified into various trigger levels for clinical alert according to RCP criteria [11], including Low, Medium, and High, etc. Our objective is to adopt the decision tree approach to directly model the vital signs and determine the risk status of individuals in the disaster-affected areas. Based on this, we can calculate the overall risk factor for each cluster, and that will help us to classify clusters needing immediate attention.

In this paper, we provide an overview of three platforms for disaster management (a) UAVs based only, (b) ground IoT network-based only, and (c) UAV-assisted ground IoT network. We propose an energy-efficient task scheduling scheme for data collection in a disaster scenario. We also present a physiological assessment of data collected by UAVs to determine the risk status of people in disaster-affected areas. Following are the main contributions of this paper:

- We present an architecture for task-based data collection by UAVs from the ground IoT network.
- We then propose an energy-efficient task scheduling scheme for data collection using UAVs from the ground IoT network to minimize the energy consumption of the UAVs.
- We also present the analysis of data and adopt a decision tree-based mechanism to determine risk status for people in the disaster-affected areas using their vital signs. This analysis will help first responders to decide an appropriate action based on the risk status of the people in disaster-affected areas. However, the methodology of action by first responders is beyond the scope of this paper.
- Finally, we present simulation results of the proposed work to evaluate its performance and compare the performance with existing work.

The rest of this paper is organized as follows. In Section 2, a brief overview of different platforms for disaster management is presented. The system model for IoT platform for disaster management using UAVs is given in Section 3. The proposed energy-efficient task scheduling scheme for data collection by UAVs from ground IoT network and analysis using a decision tree approach is described in Section 4. Simulation results are presented and discussed in Section 5. Finally, the conclusions and future research directions are provided in Section 6.

2. Platforms for disaster management

There have been research efforts in recent years for disaster management using UAVs and/or IoT systems. Here, we will provide a brief review of the application of UAVs in disaster-affected areas for SAR operations and IoT platforms for disaster management.

2.1. UAVs for disaster management

In [12], the authors identified disaster management applications of UAVs and discussed open research issues. Authors showed that UAVs, in conjunction with the wireless sensor networks (WSNs) and cellular network, have a large number of applications in disaster management scenarios. In [13], authors used a genetic algorithm to optimize the locations of UAVs to improve the coverage and throughput of the network. The algorithm to control the autonomous deployment of the UAV-assisted network is described in [14]. Simulation results and physical demonstration are used for verification. Optimization of placement of UAVs is described in [15] using a brute force algorithm. The authors showed improvement in the coverage and capacity of the network. In [16], the authors discussed the critical challenges in terms of navigation and detection autonomy. In [17], the authors proposed practical methods to assist rescuers in finding victims of natural disasters in unknown environments using lidar and infrared depth cameras. The authors presented a detection system that is insensitive to illumination change. A scheduling scheme for designating specific tasks and sensors to set of UAVs in the disaster area is proposed in [18]. Data acquired from the sensors on each UAV is sent to the cloud for use by other UAVs in the network. In [19], the authors proposed a geolocation method to search for human survivors in the disaster area. The authors used video recorders and a geolocation module mounted on the UAV. A novel method for mapping an unknown environment in case of a disaster is proposed in [20], where authors used computer vision techniques for detecting human beings. The authors also showed that quick response (QR) codes could be used for localization for the areas where the global positioning system (GPS) is not available. In [21], authors tested flying communication servers where the UAVs are equipped with WiFi, web server, WebRTC server, and a video camera. Authors proposed that the users with their smartphones will be able to exchange text chat asynchronously and communicate visually in real-time. A mobility model for self-deployment of an aerial ad hoc network in a disaster scenario is proposed in [22] based on the Jaccard dissimilarity metric. The authors tested the proposed model using software simulations that integrate the mobility of the victims and generated UAVs mobility traces based on the proposed model.

2.2. IoT for disaster management

Authors in [23] discussed key enabling technologies for IoT in the case of post-disaster response and recovery. A system of interconnected smart modules is presented in [24]. The purpose is to enable centralized data acquisition and provide a communication network to exchange useful data for monitoring purposes. A semantic early warning system is developed and tested to manage natural disasters in [25], where the authors highlighted methods to apply the semantic computing for the advancement of early warning systems. Authors in [26] used fog-computing to propose a crowdsourcing based disaster management system and presented comparison with conventional systems based on cloud computing. Authors in [27] proposed a wireless ad hoc network-based framework for disaster response. Multi-channel medium access control (MAC) is used to reduce the effect of hidden and exposed terminals and to enhance the throughput. Various passive localization systems, along with core aspects of WSNs, are discussed in [28]. The authors introduced a novel disaster management system using the WSN device-free passive localization concept. Different approaches are used to detect human motion in the disaster area, including classification, moving average, and moving variance-based detection.

2.3. UAV-based IoT network

In [29], authors studied the routing path and deployment of UAVs to acquire data from the ground sensor network. Optimal clustering of IoT devices and optimal deployment of UAVs are determined with an objective to minimize energy consumption subject to the constraint on minimum bit error rate. A comprehensive overview of UAV-based IoT services is provided in [30]. Authors anticipated UAV-based architecture for IoT services and also discussed related challenges and requirements.

2.4. Physiological assessment in disaster management

Accurately assessing the health of the population affected in a disaster scenario has been a challenging task, mainly because medical teams are not able to collect reliable data quickly from the disaster-affected area. Moreover, there are various standards to interpret the overall health of a person once the data is collected. Recently, there has been a lot of research to reduce these delays and improve the reliability of the medical data being collected from disaster-affected areas, so that the first responders can build effective response strategies with a minimum loss of life. Authors in [31] discussed a bidirectional mHealth solution to collect the data in disaster management. A device to capture and track the physiological signals of people in emergency areas is proposed using a bottom-up approach. Similarly, in [32] and [33] authors discuss the deployment of UAVs in disaster relief from a healthcare perspective and present a wide array of opportunities, including the transfer of data, transportation of medicines, vaccines, collection of blood samples. Researchers also studied the analysis of data once it is collected from disaster-affected areas. Various methods have been introduced to predict the overall risk status from the patient's vital signs. Early Warning Score (EWS) [34], modified Early Warning Score (mEWS) [35], NEWS [10], NEWS version 2 (NEWS2) are some of the common examples. These methods mainly use six or seven vital signs, including HR, S_pO_2, NBP, awRR, and Temperature. Researchers have further introduced various prediction models and compared them with these standards. Authors in [36] conducted a study of 309 patients' vital signs to predict death, hospital admission, intensive care unit (ICU) admission, and deaths in a Turkish hospital and compared the results with mEWS. Similarly, authors in [37] developed a cardiac arrest risk triage score using multivariate logistic regression to predict cardiac arrest from the physiological data. Results are compared with the mEWS system and proved that their method is more accurate than mEWS.

In a nutshell, to the best of the authors knowledge, the realistic case of data collection from a subset of cluster heads and its analysis is not explored in the literature. Hence, we target this issue in our paper and study an energy-efficient task scheduling scheme for UAVs for data collection from the ground IoT network and its analysis.

3. System model

We consider an IoT platform for disaster management using UAVs that consist of C number of clusters with N number of IoT devices in each of them, K number of UAVs, and network size S. We adopt UAVs based architecture for IoT platform, as shown in Fig. 1. It is assumed that ground IoT devices are clustered based on coverage by UAVs, and cluster heads are chosen based on residual energy. It is also assumed that cluster heads communicate their location information and UAVs share their capabilities for data collection with the controller, i.e., the monitoring center will have information about network topology of ground IoT network and capabilities of UAVs. The monitoring center can then request the UAV to collect the data from cluster heads and send it for analysis, as shown in Fig. 1. This is considered a task t. The monitoring center can request multiple tasks to the *kth* UAV. It is assumed that T_k is the number of tasks assigned to *kth* UAV. The

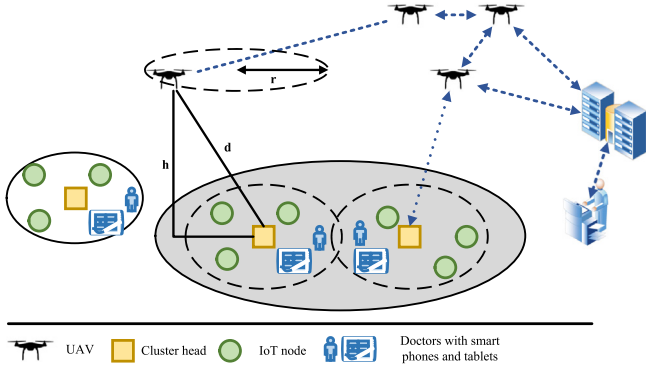


Fig. 1. IoT platform for disaster management using UAVs.

Table 2
Description of the Symbols used in this paper.

Symbols	Description
C	Number of clusters
N	Number of IoT devices on the ground
K	Number of UAVs
S	Network size
T_k	Number of tasks assigned to *kth* UAV
P_k^{UAV}	Transmit power of *kth* UAV
Φ_{kt}	Binary variable to determine if *kth* UAV can collect data for *tth* task
P_L	Path loss between cluster head and UAV
r_k	Coverage radius of *kth* UAV
E_k^{UAV}	Energy consumption of *kth* UAV
Ψ	Set of risk status classes
R_n	Risk status of *nth* IoT node associated with an individual
Θ_c	Risk factor of *cth* cluster
θ_k	Binary variable to denote scheduling of *kth* UAV
β_{kt}	Determines whether *kth* UAV can collect data from cluster head

transmit power of *kth* UAV is denoted by P_k^{UAV}. The description of the symbols used in this paper is summarized in Table 2. The *kth* UAV can collect data for task t from a cluster head if and only if it can perform that task in terms of coverage area/region and remaining/residual energy. It can be represented by Φ_{kt} as:

$$\Phi_{kt} = \begin{cases} 1, & \text{if } k\text{th UAV can collect data for } t\text{th task} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Energy consumed by UAVs depends on their route taken to cover all the assigned cluster heads, radio propagation properties, coverage range to establish communication with cluster heads, etc. The path loss can be modeled as [38]:

$$P_L(d_k) = P_{L0}(d_0) + 10\alpha \log d_k/d_0 + X_f, \tag{2}$$

where P_{L0} is the path loss from a reference location with distance d_0, d_k is the distance between *kth* UAV and cluster head, α is the path loss exponent, and X_f is the Gaussian random contributor which depicts shadowing effects.

The coverage radius of the UAV from which it can communicate with the cluster head in the ground IoT network can be calculated as:

$$r_k = \sqrt{d_k^2 - h_k^2} = \sqrt{\left(d_0 \times 10^{\frac{P_L(d_k) - P_L(d_0) - X_f}{10\Psi}}\right)^2 - h_k^2}, \tag{3}$$

where h_k is the height of *kth* UAV from the ground.

Let Γ_k be the flight duration for *kth* UAV. Then, the energy consumption of *kth* UAV can be represented as:

$$E_k^{UAV} = \left(P_k^{UAV} + P_{RF,k}^{UAV} + P_{H,k}^{UAV}\right) \times \Gamma_k, \tag{4}$$

where $P_{RF,k}^{UAV}$ and $P_{H,k}^{UAV}$ are the power consumed by *kth* UAV's RF module and hardware circuit, respectively.

Data collected by UAVs about the vital signs (physiological parameters) of people in a disaster situation is sent to the monitoring center for analysis. The collected data can be diverse and can include heart rate, blood oxygen saturation, respiratory rate, etc. Let Ψ be the set of risk status classes based on the health data collected by UAVs. Each risk status class $\psi \in \Psi$ has the following vital signs: (i) HR, (ii) S_pO_2, (iii) NBP, and (iv) awRR. We consider only limited vital signs about the health condition of individuals for illustration purposes. For data analysis, we consider data from N number of IoT devices (we assume that each IoT device is associated with one person and can have multiple sensors in it) from each cluster with R_n risk status for each individual. The value of R_n can be: 'None', 'low', 'low medium', 'medium', and 'high' based on vital signs.

We can then calculate the cluster risk factor Θ_c for each cluster using:

$$\Theta_c = \sum_{N} R_n \geq \{\text{Medium or High}\} \forall N. \tag{5}$$

In our proposed work, UAVs need to be programmed before data collection. An efficient task scheduling of UAVs along with optimal routing path, can significantly reduce its energy consumption as well as the latency. Also, low latency (which can be achieved by optimal routing path of UAVs) is of prime importance because, in a disaster scenario, administrators need to make quick decisions for which they need ultra-fast communication.

4. Energy-efficient IoT platform for disaster management

In this section, we propose a energy-efficient task scheduling scheme for data collection by UAVs from the ground network followed by the data analysis based on a decision tree to decide the risk factor for people in the disaster-affected areas.

4.1. Energy-efficient task scheduling scheme

The monitoring center can request for data collection to a controller and send a request packet for task t. The controller then calculates Φ_{kt} using (1) for all UAVs to determine capability of each UAV for data collection for task t. A *kth* UAV will be scheduled for data collection if the task t is within its coverage area. A binary variable θ_k can denote the scheduling of *kth* UAV as:

$$\theta_k = \begin{cases} 1, & \text{if } k\text{th UAV is scheduled} \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Our objective is to minimize the energy consumption of UAV by optimizing its route for data collection. The coverage range and radius of UAV are ensured. The optimization problem can be written as:

$$\begin{aligned} &\min_{\beta_{kt} \forall k,t} : \sum_{k=1}^{K} E_k^{UAV}, \\ \text{Subject to } &\beta_{kt} = \Phi_{kt}\theta_k, \\ &\sqrt{x_k^2 + y_k^2} \leq r, \\ &\beta_{kt} = \{0, 1\}, \end{aligned} \tag{7}$$

where x_k and y_k are the Cartesian coordinates of *kth* UAV at a given height h. β_{kt} determines whether *kth* UAV can collect data from cluster head.

We adopted genetic algorithm given in Algorithm 1 to solve the problem in (7). We start by generating a random population of size 100, where each population is a path of *kth* UAV to cover a set of tasks. We then continue to generate successive populations until we meet our desired criteria, i.e., the shortest path for the UAV to cover all the tasks to optimize the energy of the UAVs. To generate the new population at each successive iteration, we evaluate each member of the current population by computing its fitness value and then select the best members, called parents, based on their fitness. Here, fitness value means the distance required for the UAV to cover all the cluster heads. These parents produce children and undergo mutation during reproduction. This means we randomly swap the position of any of the two tasks to change the order in which the UAV visits them. Now we replace the members of the current population with the children, which in this case is the next generation. We continue to produce new and better generations (using selection and mutation) until we meet stopping criteria, which in our case, is 10,000 iterations. Most of the genetic algorithms also use crossover during reproduction to make changes in the offsprings. However, we did not use it here because it was increasing the complexity without improving the results. Instead, we repeated the mutation part two times to get a better offspring, i.e., a better path for the UAV.

Algorithm 1 Genetic Algorithm for Optimizing Routing Path of UAV.

```
Initialize population P(t) of different paths of UAV
while termination condition not satisfied do
    Evaluate fitness (i.e., distance traveled by the UAV) of each path in P(t)
    Select members from P(t) based on fitness
    Reproduce to create offsprings from the selected pairs
    Undergo two mutations during reproduction
    Replace members of P(t) with offspring having better fitness
end while
Decode the individual member with maximum fitness, i.e., shortest path of UAV
return the best solution
```

4.2. Analysis of data collected by UAVs

The data collected through UAVs is analyzed at the monitoring center to save the energy consumption of UAV as data analysis within the UAV would be computationally expensive. Although various types of data can be collected through UAVs, including marking the boundary of the disaster-affected area, assessment of infrastructure damaged, roads blocked, and health information of the people in disaster-affected areas. However, for the sake of simplicity, we will only discuss healthcare data in this paper because saving a human life is the most critical aspect of any disaster scenario. Some critical medical signs can help assess the general physical health of a person called vital signs. In this paper, we predict the risk status of people in disaster-affected areas using their vital signs.

4.2.1. Dataset and data preprocessing

In this paper, we are not collecting real-time vital signs data from people in disaster-affected areas; instead, we use an existing database that is provided by The University of Queensland [39]. It offers a wide range of patient monitoring data and vital signs that were recorded at the Royal Adelaide Hospital from 32 different patients. We use a subset of this data after cleaning and preprocessing that includes the following features:

- DateTime: Date and time when the data was collected from the patients using sensors.
- HR: Heart rate derived from the electrocardiogram (ECG) sensor is the speed of the heartbeat measured by the number of beats (contractions) of the heart per minute (bpm). The heart rate can vary according to the body's physical needs. However, the normal resting adult human heart rate is 60–100 bpm according to The American Heart Association states [40].

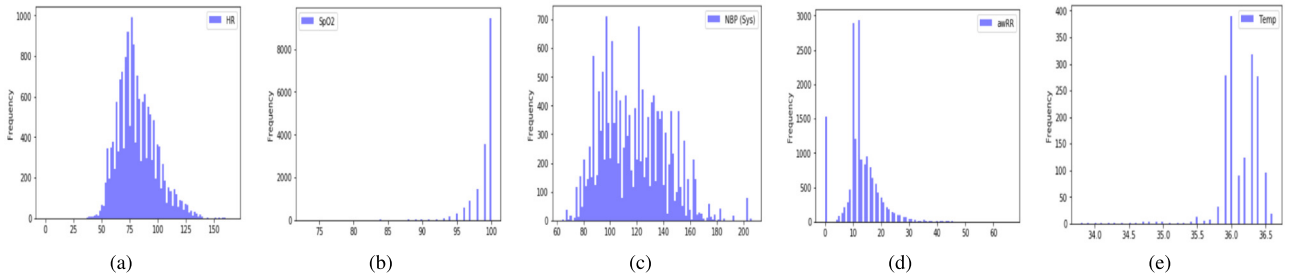


Fig. 2. Histogram of vital signs: (a) HR, (b) S_pO_2, (c) NBP, (d) awRR, and (e) temperature.

- S_pO_2: Blood oxygen saturation level derived from pulse oximetry sensors. It is the fraction of oxygen-saturated hemoglobin relative to total hemoglobin in the blood and is measured in percentage (%). Normal arterial blood oxygen saturation levels in humans are 95 to 100 percent. The human body regulates a very precise and specific balance of oxygen in the blood, and if it goes below 80% it may cause organ failure, e.g., brain or heart.
- NBP: Systolic blood pressure measured with a non-invasive blood pressure cuff. It is the maximum pressure during one heartbeat that the circulating blood exerts on the walls of blood vessels. It is a more frequent cardiovascular risk factor than diastolic blood pressure (DBP) (minimum in between two heartbeats) [41], which is why we consider it instead of DBP while calculating our risk status. It is measured in mmHg and is generally below 120mmHg [42].
- awRR: Respiratory rate derived from capnography. It is the number of breaths in one minute and is measured by counting how many times the chest rises in a minute during a rest position. The normal respiratory rate for a healthy adult at rest is 12–18 breaths per minute [43]
- Temperature: Body temperature measured using the nasal probe measured in degree Celsius.

Fig. 2 shows the histogram of the above mentioned vital signs. It can be noticed that temperature data is very sparse, i.e., there are many missing data points that can be attributed to the faulty sensors. Therefore, we only processed the remaining features, i.e., HR, S_pO_2, NBP, and awRR. We also removed the abnormal values (e.g., blood pressure = 10,000 mmHg) and filled the missing values using linear backfill, forward fill, and linear interpolation techniques [44]. We then calculated the early warning score (EWS) for each set of vital signs using the NEWS2 method mentioned in Table 1. Further, we use Pearson correlation [45] to compute the linear correlation between the features and draw a color-coded heat map shown in Fig. 3. The value of correlation ranges from −1 to +1 with zero showing non-linear correlation between the two features.

Finally, we compute the risk status from this EWS, which served as our baseline approach to training our machine learning model, i.e., decision tree classifier. We reserved 50% of the dataset for training purposes and 30% is used as a validation set. The remaining 20% is used for testing.

4.2.2. Machine learning model

After pre-processing of data, we run it through a machine learning model, i.e., decision tree. We model our problem as a multi-class classification problem because we need to predict the risk status, i.e., classify the vital signs into one of the five possible values of risk status. We use decision tree classifier and predict the risk status of a person without going through the NEWS method.

The decision tree builds a tree structure to classify the target variable by dividing the dataset into smaller subsets based on the most critical splitter in feature variables [46]. This results in the creation of a tree with decision nodes and leaf nodes. The topmost node corresponds

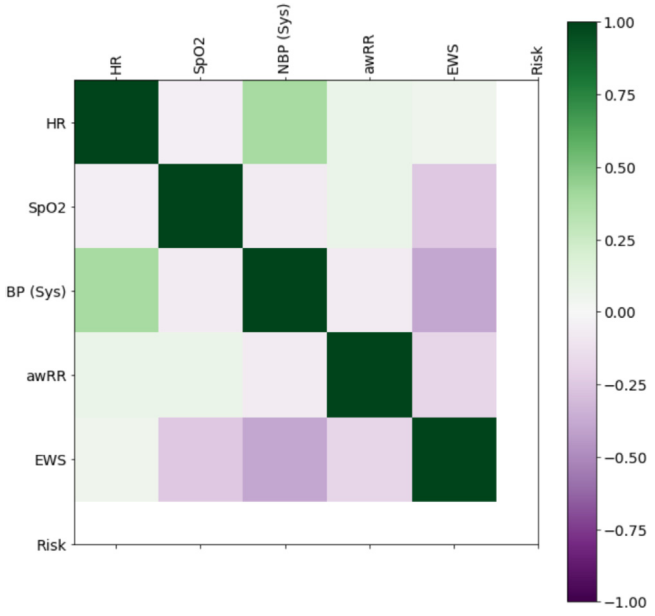


Fig. 3. Linear correlation between the features and draw a color coded heat map. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

to the best predictor called the root node. Each decision node splits into sub-nodes, which is called splitting. The final node is the terminal node, which does not split further. While forming a decision tree, we also need to remove some sub-nodes belonging to a decision node, which is called pruning. It helps to control the size of the tree and stops it from overgrowing and causing over-fitting. The decision of which node to split can significantly affect the model's accuracy. We need to find a feature that produces the most homogeneous sub-nodes. There are various algorithms to decide the splitting; here, we use the Gini index function, which tells about the purity of the nodes [47].

For the evaluation of our model, we use the confusion matrix and find the overall accuracy, along with precision and recall of each class. Since our target variable is not so unbalanced, we can choose accuracy as an evaluation metric. For a multi-class problem like the one presented in this paper, accuracy is the ratio of total true positives and true negatives to the total predicted values. However, most of the time, in a disaster scenario, we are concerned with positive outcomes. Therefore, we use the precision metric, which is the ratio of true positive outcomes to the total predicted positive outcomes. On the other hand, recall is a metric that could tell what proportion of the actual patients did we miss out, so it is also an essential factor to consider while measuring the success of a rescue mission in a disaster-affected area. Recall can be calculated by the ratio of the true positives to the total actual positives. Precision and recall are handy metrics in classification problems; however generally, it becomes a tradeoff between them. Therefore, we use the F1 score metric that can be

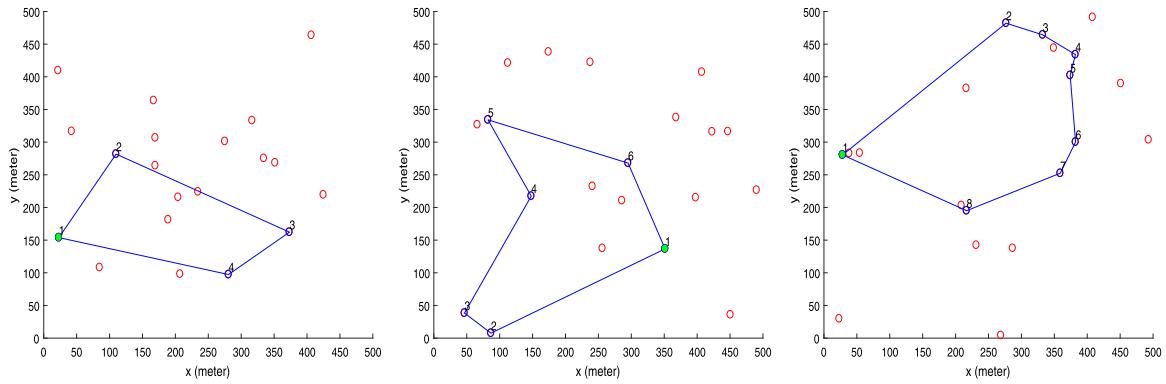


Fig. 4. Route of UAV for data collection for given number of tasks (Network size S=500 m × 500 m): (a) T_k=4, (b) T_k=6, and (c) T_k=8.

Table 3
Simulation parameters.

Parameter	Value
Network Size (S)	$500 \times 500, 1000 \times 1000, 1500 \times 1500$ m^2
Number of cluster heads (C)	10–18
Number of tasks (T_k)	0–20
Speed of UAV	30 km/h
Power ratings of UAV	500 W

interpreted as a weighted average of precision and recall. It ranges from 0 to 1, with 1 being the best score. We calculate it using the following formula:

$$F1 = \frac{2 \times (\text{precision} \times \text{recall})}{\text{precision} + \text{recall}}. \quad (8)$$

We also use the k-fold cross-validation to evaluate our model in which we divide the data into k number of folds. Each fold is kept for testing while training is done on the remaining folds. This ensures that each part of the data is being used for training and testing at least once. Evaluation metrics are measured for each fold, and finally, their average is taken as a final metric.

5. Simulation results

In this section, we evaluate the performance of our proposed energy-efficient scheme for data collection in disaster scenarios and data analysis based on the dataset [39]. It is important to note that we are assuming that UAVs will collect data similar to the given in the dataset in the disaster situation. For the sake of simplicity, we consider one UAV in our simulations. However, the proposed scheme can be extended to have multiple UAVs, which can certainly enhance the performance of the IoT platform. We have distributed our sensor nodes randomly in a square area. Detailed simulation parameters are given in Table 3.

5.1. Energy-efficient task scheduling scheme

Fig. 4 shows the distance traveled by the UAV for different number of tasks assigned. It is clear that increasing the number of tasks for the UAV increases its total distance traveled, hence consume more energy. We assume the same network size in all three experiments. As the number of tasks increases, we see that our model becomes closer to the traditional approach where there is no selection of tasks. Hence, we conclude that our proposed model with the right choice of tasks reduces the overall energy consumption of the UAV.

Fig. 5 shows the energy consumption of the UAV against the number of tasks at various network sizes for a given number of cluster heads, i.e., 20. We compare the results of our proposed model with the traditional approach where there is no selection of tasks and the UAV has to hover over all the cluster heads. We see that by increasing the number of tasks for any given network size, the energy consumption of UAV increases almost linearly. We conclude that we can conserve a significant amount of energy of the UAV with proper selection of tasks, and hence increase its flight time when compared with the traditional approach.

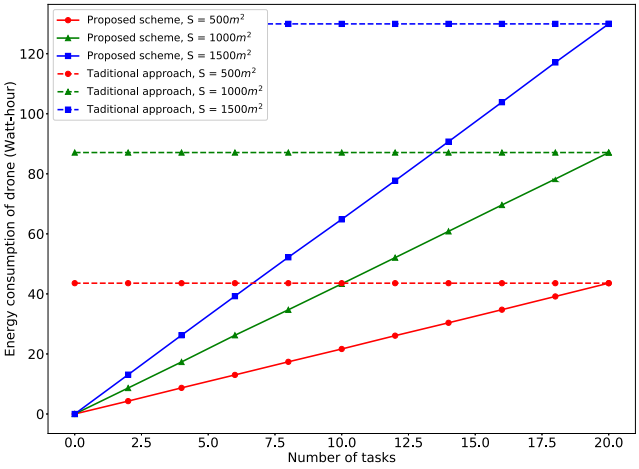


Fig. 5. Energy consumption versus number of tasks for different network sizes.

Fig. 6 shows the effect of the number of cluster heads on energy consumption for different network sizes. We see that the total number of cluster heads does not affect the energy consumption, rather it is the number of tasks that matter for the flight time of the UAV. Therefore, for given network size, energy consumption is the same for any number of cluster heads. However, network size has a significant effect on energy consumption, and a positive linear relationship is found among them. We also studied the effect of a number of tasks, and the results reiterate the findings of Figs. 4 and 5, i.e., energy consumption increases linearly with the number of tasks, and is maximum when the number of tasks becomes equal to the number of cluster head (traditional approach).

5.2. Analysis of data collected by UAVs

We consider a database of 16,671 vital signs datasets provided by University of Queensland [39]. The distinct values for each vital sign with mean and standard deviation is given in Table 4.

We analyze this data as described in Section 4.2 and the results are presented in Table 5. It shows the predicted versus true results

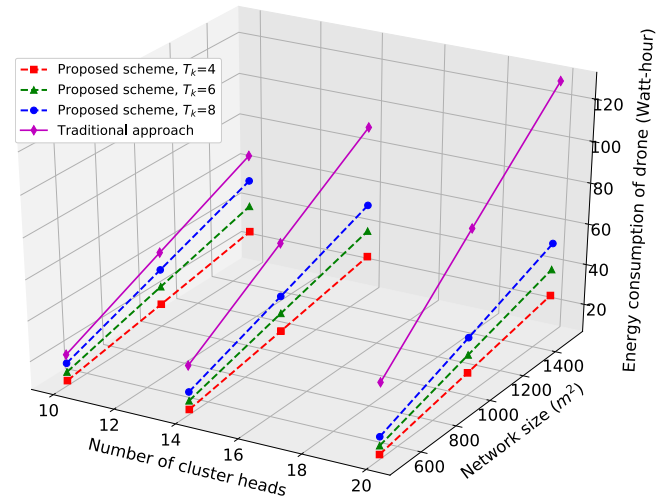


Fig. 6. Energy consumption of UAV versus number of cluster heads and network size.

Table 4
Details of vital signs data.

Score	HR	S_pO_2	NBP	awRR
Distinct	118	28	117	58
Mean	81.49	98.98	117.47	12.68
Std. deviation	17.07	1.95	24.05	6.49

for the risk status classified as 'None', 'low', 'low medium', 'medium', and 'high' based on vital signs using the decision tree model. We have predicted 27 people with high-risk status. Similarly, we can predict the risk status for the vital signs data collected by UAVs from other clusters and find the overall risk factor of each cluster. Based on the risk factor, we can decide clusters that need immediate attention. Table 6 shows the evaluation metrics for our model. We use three metrics: precision, recall, and F1-score, which are calculated for each risk status. Our adopted decision tree model has predicted the risk status with reasonable high precision, recall, F1-score. High precision means that the number of people we have identified in high-risk status is actually in high-risk status. In our scenario, this means we can deploy doctors' teams in the high-risk cluster to save precious human lives. Similarly, recall shows that the proportion of people that are in high-risk status is correctly classified as high risk. Reasonably high values of F-score (a combination of precision and recall (8)) shows the efficiency of the adopted decision tree model for predicting people in high-risk to deploy doctors team in that cluster.

We further evaluate our model using k-fold cross-validation using the value of $k = 5$ folds. We choose $k = 5$ based on the number of data points (16,600) in our data, and also, it is the default value recommended by the scikit-learn library [48]. This means that we set aside 20% of the data points for testing and train using the remaining 80% of the data. This process is repeated for each of the next 20% of the data (also known as folds). If we consider a high value of k, e.g., $k = 30$, this would mean that we have only 3.3% of the data for testing, which would be insufficient to reach a reliable conclusion. On the other hand, if we take a very low value of k, e.g., $k = 2$, it would mean that we have set aside only 50% of the data points for training, which again would not be sufficient for proper training of a model. Hence, we decided to go with a value of $k = 5$. We found the evaluation metrics for each fold, and the average of five folds is presented as the final metric. For each fold, we got accuracy values ranging from 82.6% to 95.3% with a mean accuracy of 87.2%. The standard deviation of the accuracy is 0.04.

Table 5
Confusion matrix: predicted versus actual values.

	Predicted				
True	None	Low	Low medium	Medium	High
None	468	23	19	15	0
Low	52	1607	85	28	1
Low medium	20	46	684	11	1
Medium	5	1	3	226	3
High	1	1	2	1	22

Table 6
Evaluation Metrics.

Risk status	Precision	Recall	F1-score
None	0.86	0.89	0.87
Low	0.95	0.90	0.93
Low medium	0.86	0.89	0.88
Medium	0.80	0.95	0.87
High	0.82	0.80	0.81

6. Conclusions

Energy consumption is a serious threat to the performance of IoT-platform for disaster management using UAVs. Energy consumption can be minimized by optimizing the flight route of UAV to collect data from the ground IoT network. This paper presented an overview of existing platforms for disaster management based on UAVs or IoT. We also proposed an energy-efficient task scheduling scheme for collecting data from the ground IoT system to minimize the energy consumption of UAVs. We formulated a mathematical framework to minimize the energy consumption of UAVs subject to the constraints on the coverage area of UAVs within which it can communicate with the cluster head. We used the genetic algorithm to find the optimal route for the UAVs. The performance of a energy-efficient task scheduling scheme is analyzed through simulation results. We studied the impact of the number of tasks, the number of cluster heads, and network size on the energy consumption of UAV. It is concluded that efficient scheduling of UAVs can notably reduce the energy consumption when compared with the traditional approach for data collection by UAVs from ground IoT network. We have also analyzed vital signs data collected by UAVs and identified the health risk status of people in the disaster-affected area into 'none', 'low', 'low medium', 'medium', and 'high'. We adopted the decision tree approach to predict the risk status and achieved high precision, recall, and F1-score ranging from 0.81 to 0.93.

As a potential future work, our proposed scheme can be extended to use multiple UAVs, which can further optimize the energy consumption by proper task allocation for each UAV. Also, communication protocols can be improved between UAVs and cluster heads as well as UAVs to the monitoring center to enhance the connectivity. Another interesting investigation can be the allocation of tasks based on risk factors of each cluster, i.e., the first time we can collect data from UAVs, and after analysis of collected data, we can prioritize the route of UAVs to clusters with high-risk factor.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Waleed Ejaz: Conceptualization, Investigation, Methodology, Project administration, Resources, Validation, Writing - original draft, Writing - review % editing. **Arslan Ahmed:** Data curation, Formal analysis, Software, Writing - review & editing. **Aliza Mushtaq:** Conceptualization, Validation, Resources, Writing - review & editing. **Mohamed Ibnkahla:** Supervision, Project administration, Writing - review & editing.

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